

## **Project Objective**

The objective of this project was to clean and prepare a raw sales dataset for accurate analysis and reporting. The dataset contains transactional sales information related to products, referral sources, payment methods, coupon usage, customer purchases, and revenue performance.

## **Dataset Description**

The sales dataset includes the following fields:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

## **Cleaning Techniques Used**

Several data cleaning techniques were applied to improve data quality and prepare the dataset for analysis:

- Removed duplicate transaction records
- Corrected inconsistent date formats
- Standardized column data types
- Corrected formatting inconsistencies
- Replaced missing/null values
- Validated revenue and quantity fields
- Improved dataset consistency and structure

## **Missing Value Handling**

Missing values were identified in the CouponCode column. Appropriate replacements were applied to maintain dataset completeness and ensure analytical reliability.

## **Formatting Corrections**

The following improvements were applied:

- Standardized date formatting
- Corrected currency and numeric fields
- Updated text formatting consistency
- Verified proper column data types
- Ensured values were analysis-ready

## **Data Validation**

The cleaned dataset was reviewed to confirm:

- No duplicate transaction records remained
- Numeric columns contained valid values
- Revenue calculations were accurate
- Missing values were resolved
- Column formatting was standardized

## **Result**

The dataset was successfully transformed into a clean, structured, and analysis-ready format suitable for:

Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA)
- SQL querying
- Power BI dashboard creation

The cleaned dataset now supports accurate analytical insights and professional reporting.

## **Tools Used**

- Microsoft Excel

## **Conclusion**

This project demonstrated practical data cleaning techniques required in real-world data analytics workflows. Proper cleaning and preparation improved data accuracy, consistency, and reliability, enabling the dataset to be effectively used for further business analysis and visualization projects.